



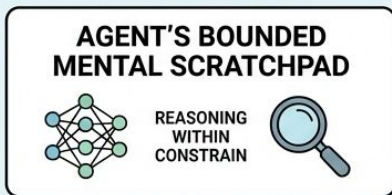
U.S. General Services
Administration

Chapter 5.9: Working Memory

US General Service Administration
AI Community of Practice - March 2026

Context Windows

UNDERSTANDING THE BOUNDS OF AGENT COGNITION



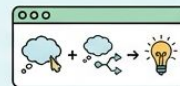
CONTEXT WINDOW IS WORKING MEMORY FOR LLMs



TOKENS COMPETE

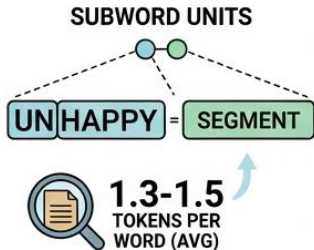


BIGGER WINDOWS $\neq$ BETTER THINKING
'Large but Cluttered' mixed $\neq$ 'Structured and Efficient' deliver clear insight

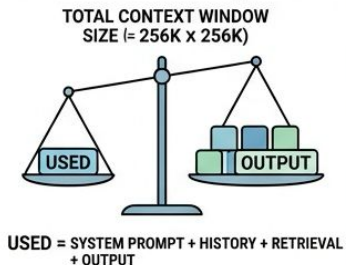


MANAGING THE LIMITED CONTEXT WINDOW BUDGET

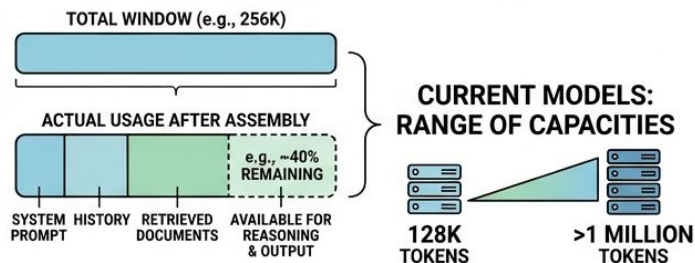
WHAT IS A TOKEN?



TOKEN BUDGET EQUATION

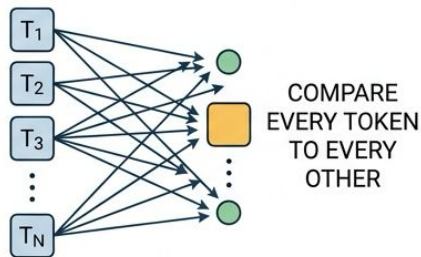


SHARED CAPACITY CHALLENGE

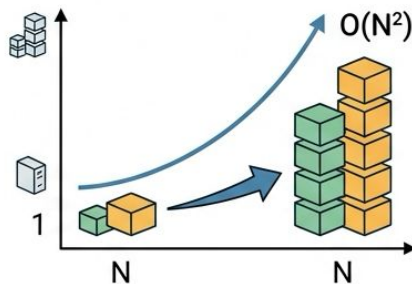


Why Hard Limits on Context Windows?

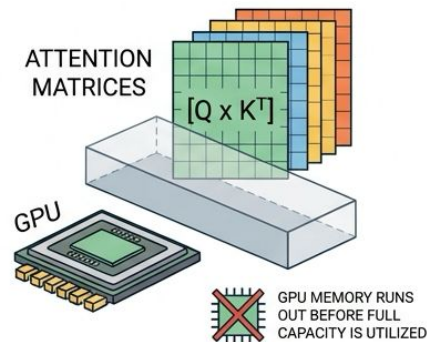
TRANSFORMER ATTENTION MECHANISM



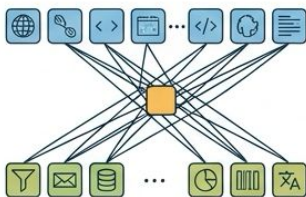
$O(N^2)$ SCALING



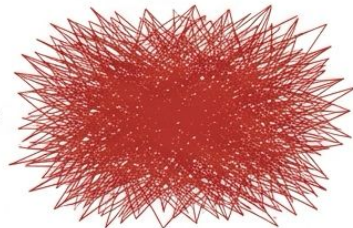
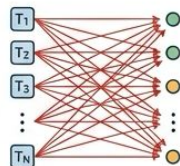
GPU MEMORY FILLS QUICKLY



CONTEXT LENGTH vs. COMPARISONS

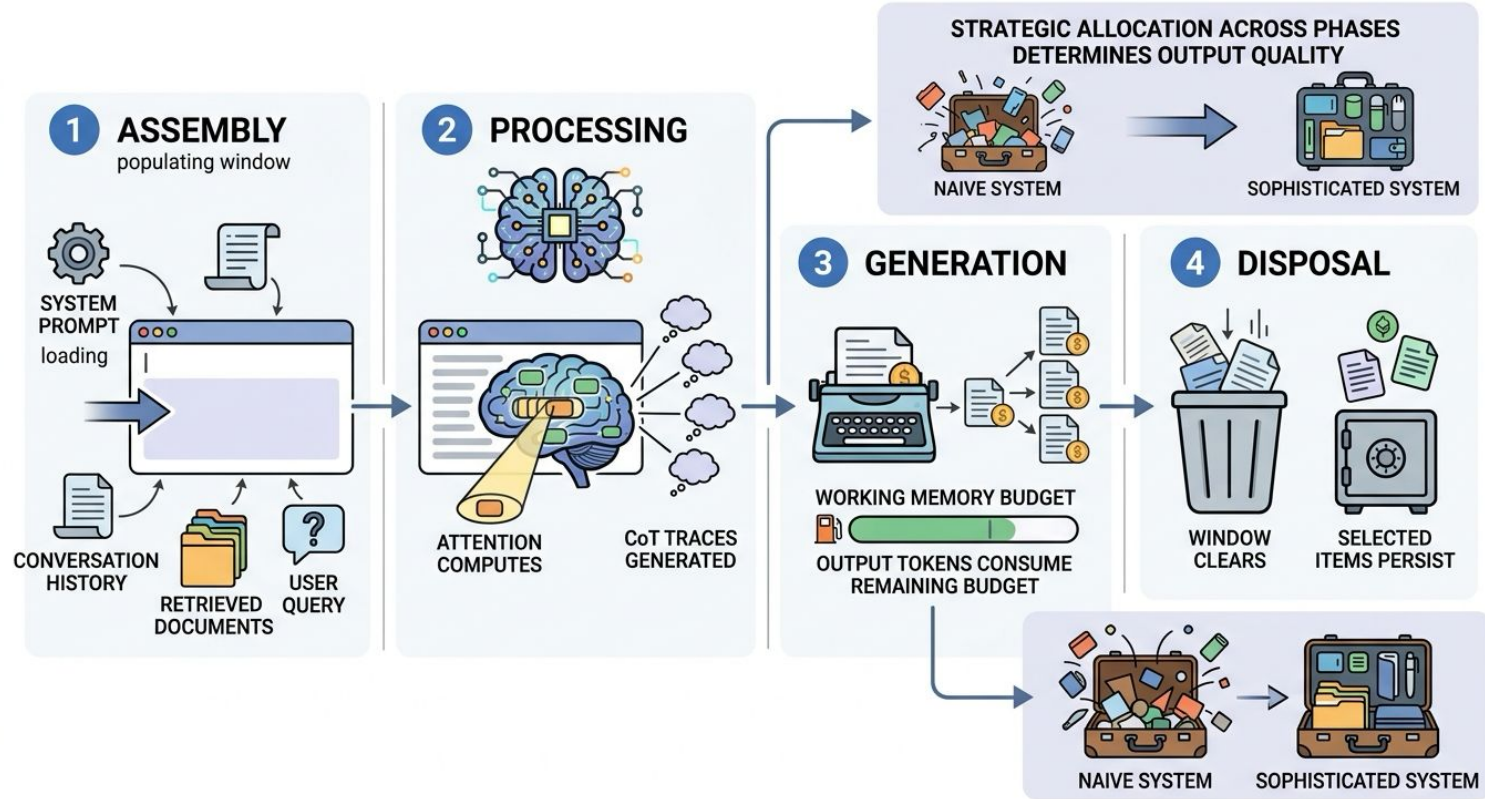


e.g.,
100K
TOKENS

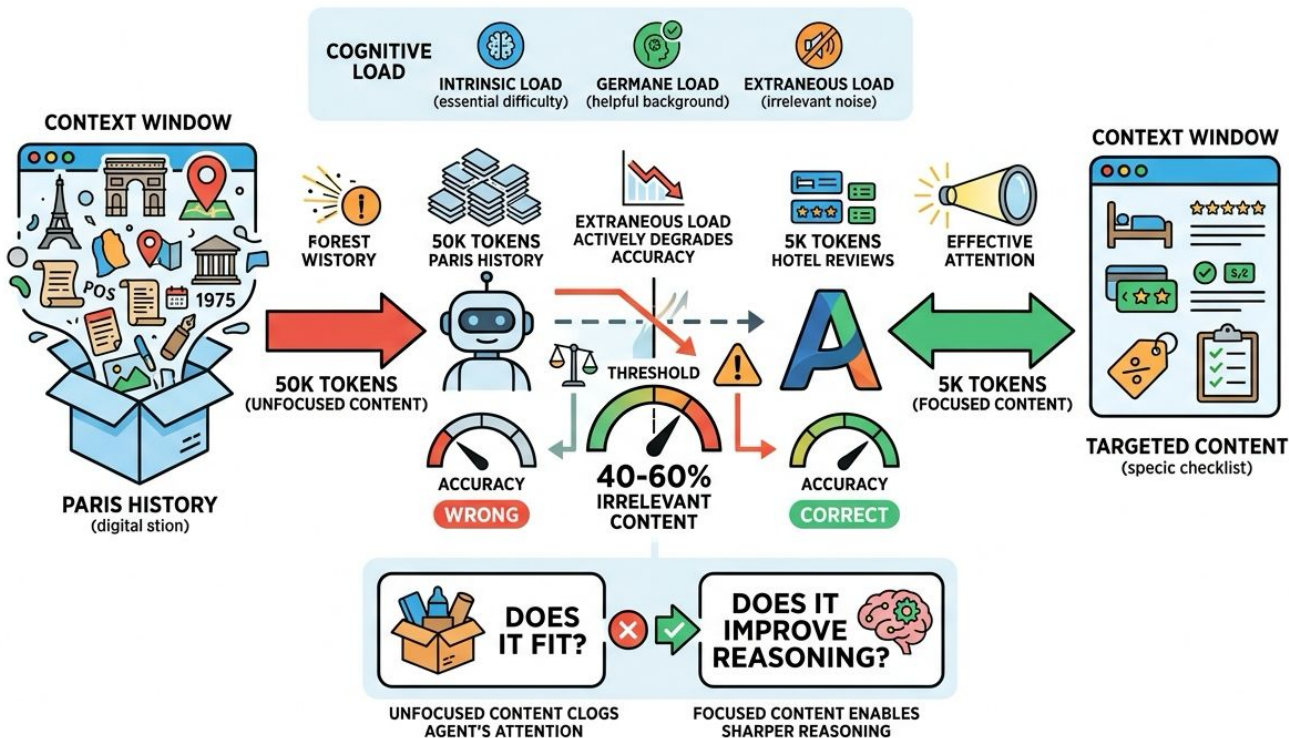


~10 BILLION
COMPARISONS PER
ATTENTION LAYER

The Working Memory Lifecycle

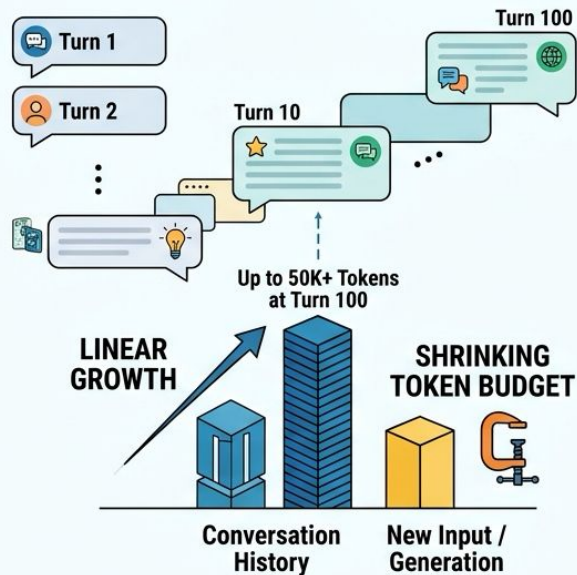


Cognitive Load -- Why Filling the Window Hurts



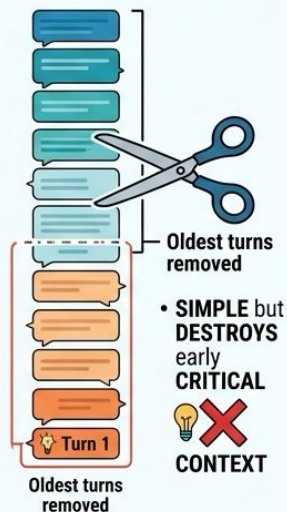
Multi-Turn Conversations

THE ACCUMULATION PROBLEM

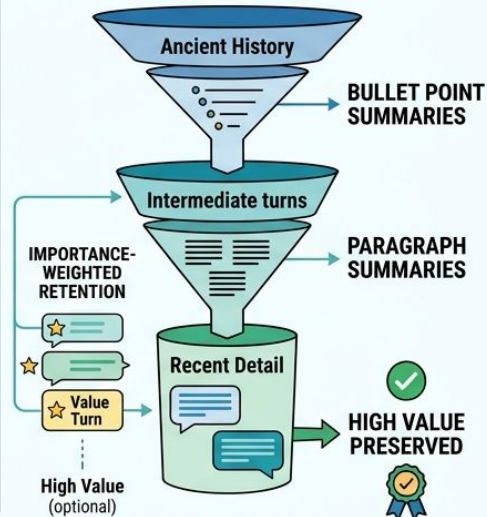


STRATEGIES FOR HISTORY MANAGEMENT

SIMPLE TRUNCATION

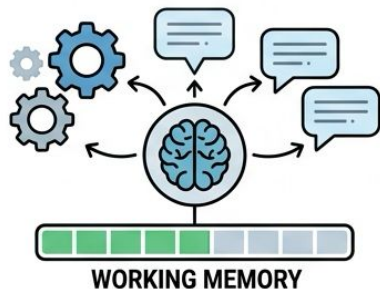


HIERARCHICAL COMPRESSION

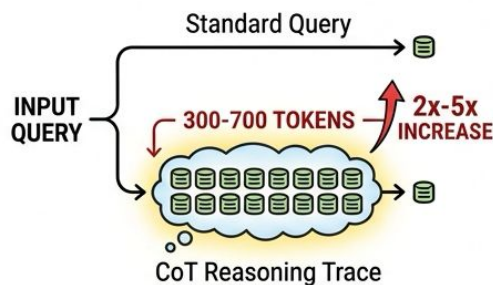


Trading Tokens for Accuracy

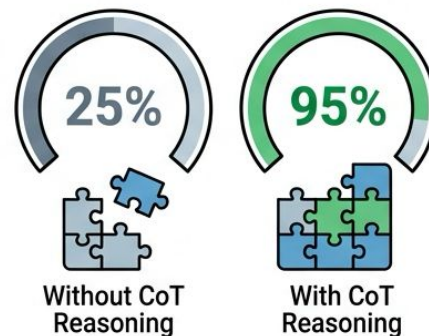
CoT Externalizes Reasoning Steps into Working Memory



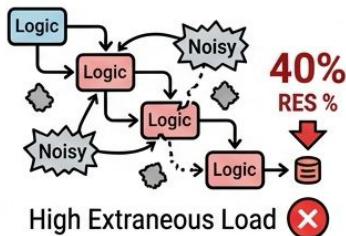
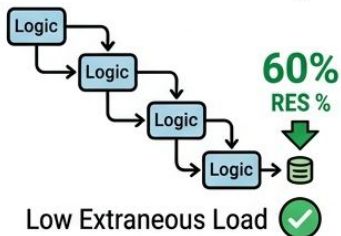
Adds 300-700 Tokens Overhead per Query (2x-5x Increase)



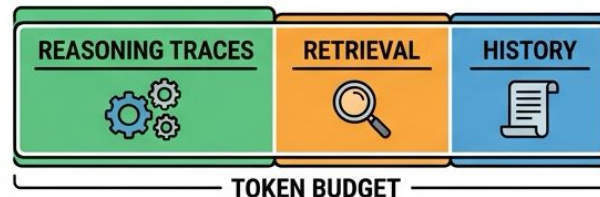
Delivers 10-30% Accuracy Gains on Complex Tasks



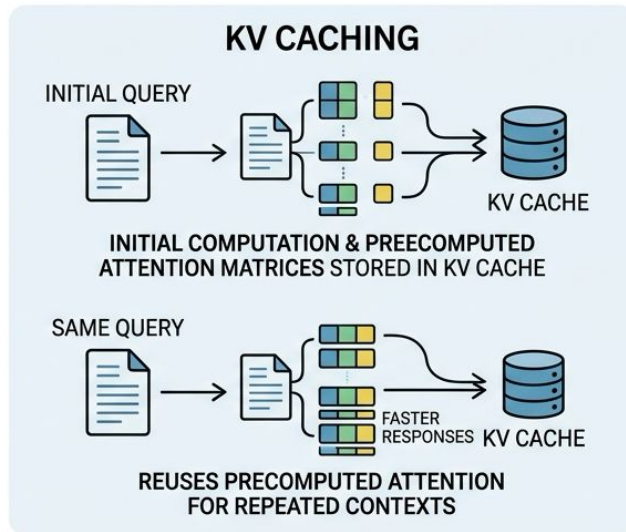
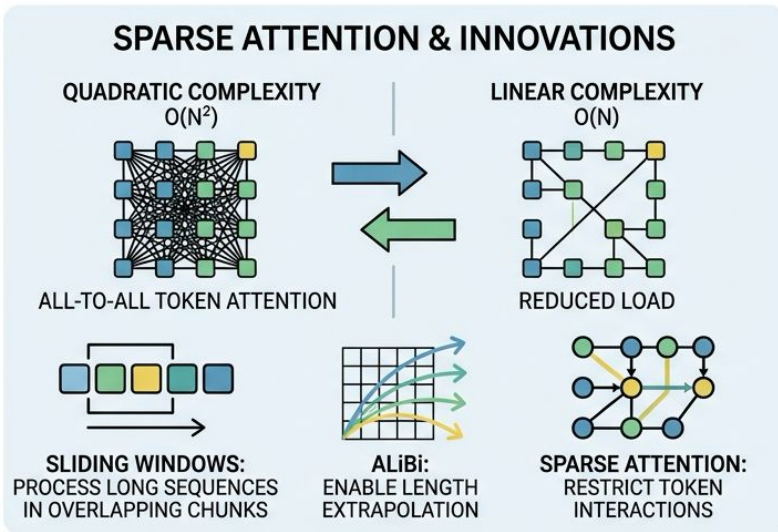
Multi-Hop Reasoning Degrades Sharply Under High Extraneous Load



Budget Must Explicitly Reserve Space for

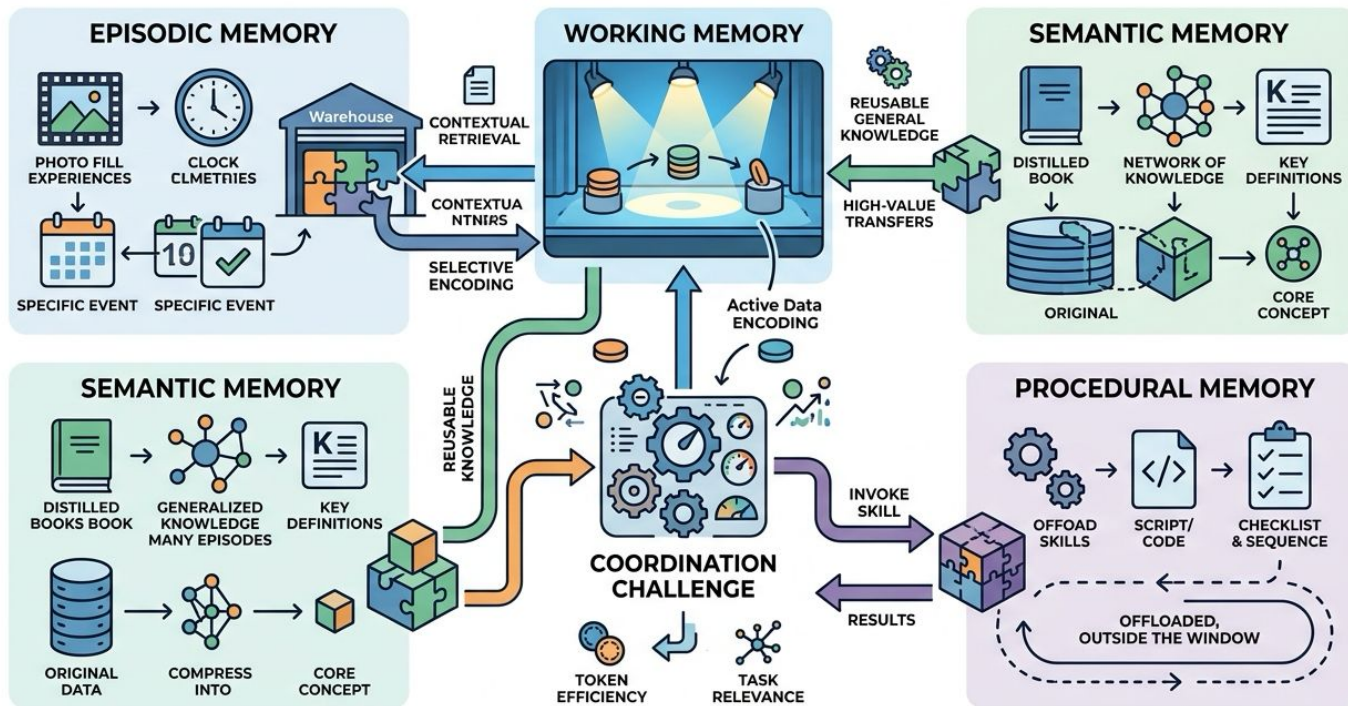


Sparse Attention and KV Caching



*Strategies adapted from leading AI research

Integration with Long-Term Memory Systems



Real-World Patterns -- Production Working Memory

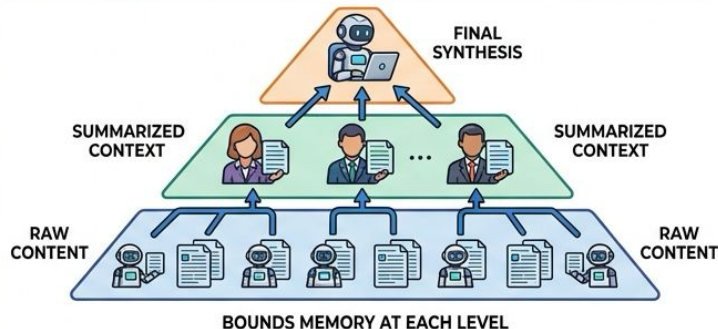
- Customer support: hierarchical compression maintains 89% completion at turn 50
 - Code analysis: curated 30K tokens beats full 200K codebase (81% vs 34%)
 - Legal review: multi-pass structured processing hits 91% vs 68% single-pass
 - Literature synthesis: three-pass approach scores 8.1/10 vs 5.2/10
 - Token curation consistently outperforms token volume
- 

Multi-Agent Working Memory Coordination

SHARED CONTEXT POOL



HIERARCHICAL AGGREGATION



MESSAGE-PASSING SHARES MINIMAL CONTEXT



COORDINATOR-MANAGED TOKEN BUDGETS



Key Takeaways

- Working memory is bounded, ephemeral, and shared across components
 - Manage cognitive load, not just token count
 - Hierarchical compression and lazy retrieval are production essentials
 - Integration with long-term memory multiplies effective capacity
 - Next chapter: utility-based decision-making (Chapter 5.10)
- 